FoodSprint

AI-POWERED FOOD ORDERING SYSTEM

AI Smart Food Ordering and Diet Recommendation System

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SLIDE 2

Introduction of Project

"A location-aware food ordering platform built for Telangana — powered by Flask and AI."

FoodSprint is a Flask-based food ordering web application designed for users in Telangana. It enables users to discover nearby restaurants, browse menus, add items to a live cart, make UPI QR-based demo payments, and track their orders in real time — all from one seamless web interface.

Nearby Restaurant Discovery

Uses OpenStreetMap and the Overpass API to detect the user's location and display nearby restaurants on an interactive Leaflet map — with a Hyderabad fallback if location is unavailable.

Full Ordering Workflow

Users can browse menus, add items to a live cart, proceed to checkout, simulate UPI QR-based payments, and track the step-by-step order lifecycle — from placement to delivery.

AI Diet Recommendations

An AI-based diet recommendation engine suggests health-friendly food options based on user dietary preferences and personalized suggestions derived from past order history.

SLIDE 3

Objectives

Smart Ordering Platform

Build a complete, smart food ordering platform that streamlines the end-to-end experience from discovery to delivery.

Location-Based Discovery

Show nearby restaurants using the user's live GPS location with an OpenStreetMap + Leaflet-powered interactive map.

Menu, Cart & Payment

Provide full menu browsing, live cart updates, checkout flow, and UPI QR-based demo payment simulation.

Additional Objectives

Admin & Staff Management: Build an admin dashboard and staff-based order status update workflow for restaurant operations.
AI Diet Recommendations: Integrate an AI-based logic engine to recommend diet-friendly food tailored to user preferences and history.
Deployable Full-Stack App: Deliver a production-ready, deployable full-stack web application hosted on Railway with version control on GitHub.

SLIDE 4

Stakeholders

Customers / Users

- Browse and discover nearby restaurants
- Order food, make UPI payments
- Track orders and receive AI diet tips

Project Team / Developers

- Design, develop, and deploy the application
- Integrate AI features and map services
- Maintain codebase on GitHub

Restaurant Admin

- Manage restaurant listings and menus
- Monitor incoming orders via dashboard
- Update item availability and pricing

Evaluators / Faculty

- Assess the project for academic evaluation
- Review AI tool usage and code generation
- Validate deployment and functionality

Staff / Delivery Handlers

- Receive order notifications
- Update step-by-step order lifecycle status
- Coordinate delivery and fulfilment

SLIDE 5

Technologies, Tools & Software

Frontend

HTML, CSS, JavaScript — for responsive UI, interactive pages, and live cart behavior.

Backend

Python Flask — lightweight web framework powering all routes, logic, and API endpoints.

Database

SQLite (development) / PostgreSQL (production) with SQLAlchemy ORM for data modeling.

Maps & Location

OpenStreetMap + Leaflet.js for interactive maps; Overpass API for nearby restaurant discovery.

Payment

UPI QR simulation for demo checkout; no real transaction — purely front-end simulation.

Deployment & DevOps

Railway for cloud deployment; GitHub for version control; ChatGPT / Codex as AI coding tools.

SLIDE 6

AI Prompts Used to Build FoodSprint

A clean summary of the four core prompts that shaped the FoodSprint application, from the base project to admin tools, AI recommendations, and map-based restaurant discovery.

```
rompt 1: Full Base Project

ext
d a full-stack, production-ready food delivery w
ication using Flask named "FoodSprint".

irements:
dern Swiggy/Zomato-style UI
er signup, login, logout
staurant listing with cards
ique menu for each restaurant
d to cart, remove from cart, quantity update
eckout flow
der system
sponsive design
Lite/PostgreSQL support
ean modular code structure
 not use static boring UI, make it modern and po
```

Full Base Project

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rompt 2: Admin, Staff, Payment, and Order Tracking

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nce my existing FoodSprint app without removing any feat

ngle admin system
min dashboard
x 3 staff accounts with staff_id like STF001
aff login and staff dashboard
aff can update order status only
der status flow:
ACED
EPARING
T_FOR_DELIVERY
LIVERED
I QR payment simulation
min can set UPI ID
ymnt confirmation should generate transaction ID
er order tracking timeline page
cure password hashing
ssion-based authentication
```

Admin, Staff, Payment, and Order Tracking

```
Prompt 3: AI Recommendation System

text
ance my existing FoodSprint app with AI-based recommendatio
tures.

:
iet recommendation system for
eight loss
uscle gain
alanced diet
egetarian
how recommended foods and foods to avoid
how calories
dd smart tags like:
ealthy
igh Protein
ow Calorie
opular
rack user order history
ecommend foods based on past behavior
eep the logic simple, explainable, and modular
o not break existing app features
```

AI Recommendation System

```
mpt 4: Map-Based Restaurant Discovery with OpenS

t
e my existing Flask-based FoodSprint app by addi
 restaurant finder without using Google Maps API

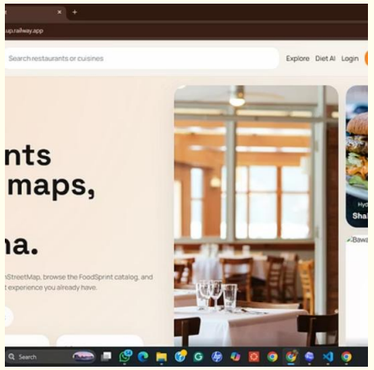
ements:
OpenStreetMap tiles
Leaflet.js on frontend
Overpass API on backend
ct browser location
ack to Hyderabad if location permission is deni
 nearby restaurants on the map
marker clustering
king a marker should show restaurant details
filters for city, type, keyword, distance, ratin
g
 all existing FoodSprint features working
```

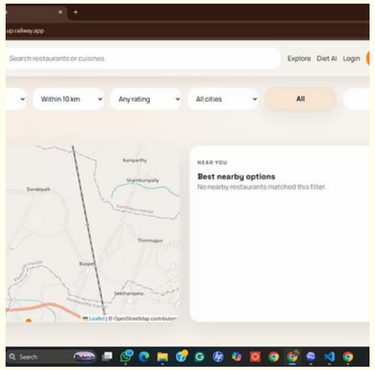
Map-Based Restaurant Discovery

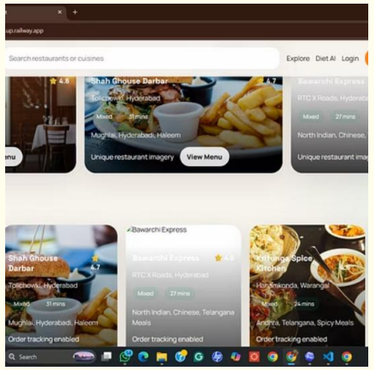
SLIDE 7

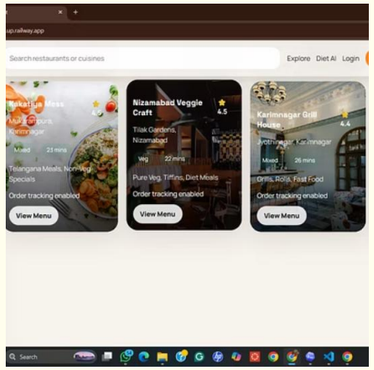
Website Screenshots

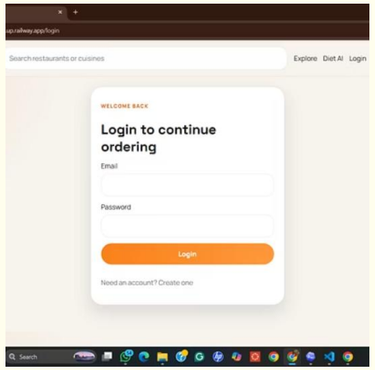
Realistic screenshots of the live FoodSprint web application, showing the key user and admin flows end to end.

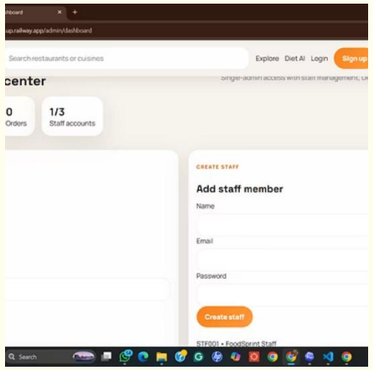


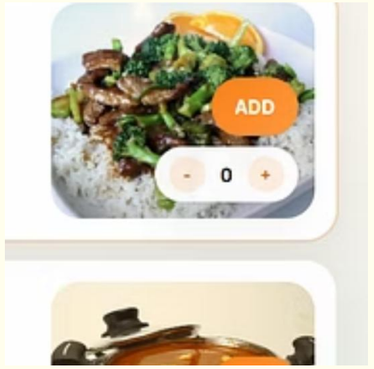


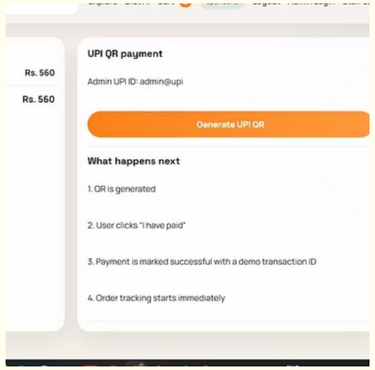








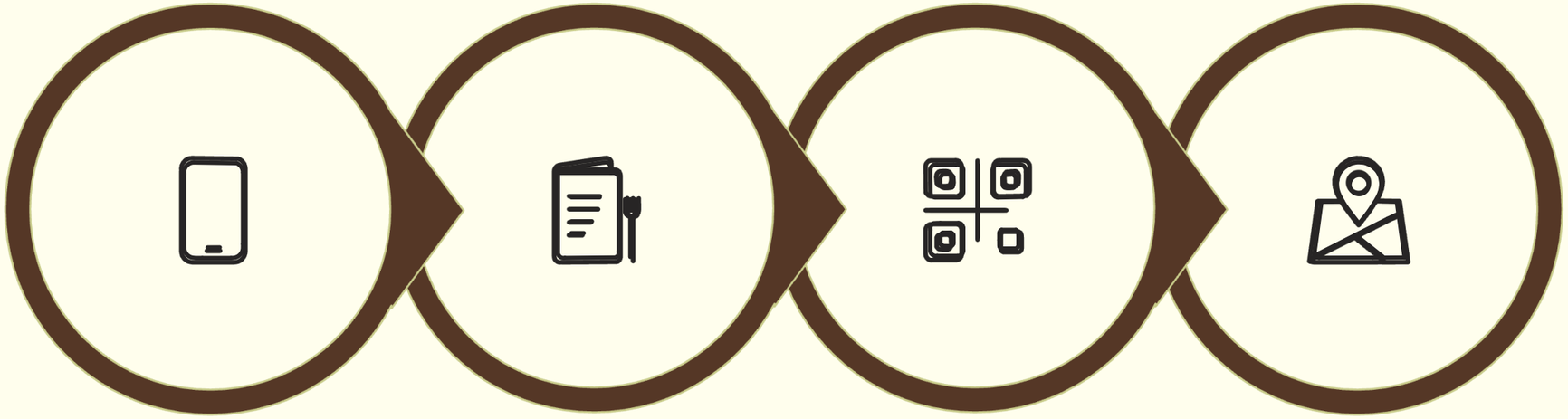




SLIDE 8

Workflow Flowchart

The complete user journey from app open to order tracking — including location detection, payment, and staff-based status updates.



Open App *Select Restaurant* *Confirm Payment* *Track Order*

The workflow is fully automated from discovery to delivery. The Overpass API powers restaurant detection, while the staff panel drives order status updates that users can track in real time.

SLIDE 9

Front-End View of the Web Application

Screenshots of the key pages across the FoodSprint application — from homepage to admin dashboard.

Homepage

Landing page with app branding, location prompt, and quick-access navigation to restaurants and cart.

Map View

Interactive Leaflet map displaying nearby restaurants via Overpass API with clickable location pins.

Menu & Cart

Restaurant menu page with item listing, add-to-cart buttons, and live cart counter updating in real time.

Checkout & Payment

UPI QR code payment simulation screen with order summary, total amount, and confirmation flow.

SLIDE 10

References & Project Links

Documentation References

Maps & APIs

- OpenStreetMap — openstreetmap.org
- Leaflet.js — leafletjs.com
- Overpass API — overpass-api.de

Backend & Database

- Flask — flask.palletsprojects.com
- SQLAlchemy — sqlalchemy.org
- Python — python.org

Deployment & Version Control

- Railway — railway.app
- GitHub — github.com

Project Quick Access

🌐 **Live Website:** https://web-production-d28ec.up.railway.app/

💻 **Source Code:** https://github.com/laxmispandana/Food_Sprint

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